

Whac-A-Mole: Exploring Virtual Reality (VR) for Upper-Limb Post-Stroke Physical Rehabilitation based on Participatory Design and Serious Games

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ABSTRACT

This paper describes a Human-Centered Design process aimed at understanding how Virtual Reality (VR) can assist in post-stroke physical rehabilitation. Based on insights from stroke survivors and healthcare professionals, a serious game prototype is proposed. We focused on upper-limb rehabilitation, which inspired the game narrative and the movements users must perform. The game supports two modes: 1- normal version - users can use any arm to pick a virtual hammer and hit objects; 2- mirror version - converts a traditional approach to VR, providing the illusion that the arm affected by the stroke is moving. These were evaluated through a user study.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction Paradigms—Mixed/augmented reality;

Human-centered computing—Human computer interaction (HCI)—HCI Design and Evaluation Methods—User Studies

1 INTRODUCTION

Stroke is one of the leading causes of disability and death that affects the world population. It can be defined as an ischemic or hemorrhagic death of the cells in the brain, spinal cord or retina, with clinical or imaging evidence of permanent injury [6].

Rehabilitation is a relevant part of the life of stroke survivors. Much of the recovery can be achieved in the acute phase (up to two weeks) and subacute phase (up to six months). As for the chronic phase (more than six months), most techniques focus on maintaining muscle strength and avoid deterioration and weakness. Therefore, rehabilitation activities need to be progressive, dynamic and goal-oriented, with the objective of attaining the best possible quality of life for the survivor and their family [3].

Nevertheless, maintaining motivation during the rehabilitation process is challenging for survivors, particularly using conventional laboratory-based techniques. Social isolation is also a common problem for stroke survivors, often resulting in depression and anxiety. In addition, the pandemic constraints also affected the rehabilitation process. As a consequence, survivors experienced a negative impact on their treatment due to a shortage of rehabilitation activities [5].

In this vein, Virtual Reality (VR) technologies may be used as a complementary type of therapy for stroke rehabilitation, having the potential to be used in real-life settings (e.g., at home) and to increase motivation for rehabilitation exercises, as well as to improve the likelihood of survivors continuing their treatment over time [1, 2, 4, 7].

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2 PARTICIPATORY DESIGN WITH STROKE SURVIVORS

To inform the design of a VR serious game for physical rehabilitation in real-life settings, allowing survivors to exercise their upper limbs in more familiar environments, a participatory process was conducted based on (see Figure 1): 1- online questionnaire with 20 chronic phase survivors; 2- focus group with 8 survivor's, 1 physician and 3 engineers. The main goal was to understand survivors' challenges, needs and expectations in order to contribute to their quality of life and independence when doing daily, common tasks.

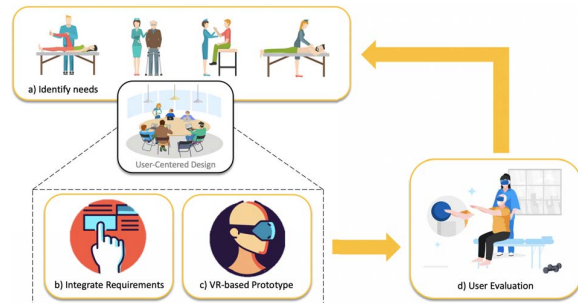


Figure 1: Methodology adopted to bring stroke survivors into the design of a VR-based application: a) focus group and online questionnaire to identify the main needs based on storyboards; b) integration of requirements; c) creation of a VR-based prototype; d) user evaluation.

The online questionnaire was made available to a non-governmental organization for stroke survivors. It had questions regarding survivors preferences in leisure and exercising activities, how much time they spend on these, what activities they would like to do, and their experience with video games and VR. Results showed a trend towards simulation and action games. This is not a surprising finding, but it shows there is potential for replacing this time with VR games. For the focus group, storyboards were used to introduce the concept of VR, and serious games, its usage in rehabilitation and potential benefits. Also, what types of experiences have could be more successful to keep survivors motivated. It gave the opportunity to gather information about survivors' opinions, their concerns, challenges and needs, as well as how they felt the technology should be used. For example, negative feedback, which is usually present in video games as an important tool for challenging users, can be a demotivator for depressed patients. Some type of leniency or positive reinforcement should be considered instead. We also found that in the chronic stage, survivors are more willing to engage with experiences they can easily relate to real-world activities, e.g., walking, traveling, swimming, playing tabletop games, cooking, or gardening. Conventional approaches fail to take this into consideration, leading to the lack of interest over time. VR appears as promising for such scenarios. In fact, survivors were quite interested in the possibility of having VR-solutions at home, which may contribute to perform rehabilitation in less intrusive, more natural and ecological settings, in contrast with typical laboratory environments.

3 WHAC-A-MOLE: VIRTUAL REALITY SERIOUS GAME

Based on the insights obtained through the questionnaire and the focus group, an immersive VR serious game for upper-limb rehabilitation was developed (see Figure 2), taking inspiration from the traditional whac-a-mole pastime as the game narrative. The user is immersed in a virtual environment with the perception of being physically present there through the aid of image and sound stimuli. This way, users can visualize their hand doing the movements, while they stimulate the corresponding area of the brain that was affected. These movements are considered useful in the scope of upper limb rehabilitation. Two modes are available: 1- normal version, i.e., users can use any hand to pick a virtual hammer and hit the moles; 2- mirror version, i.e., converts mirror therapy to a VR setting, thus providing the illusion that the arm affected by the stroke is moving. Plus, three levels of difficulty exist, changing the number of moles and the frequency of appearance, allowing survivors to personalize their experience. Also, there are no penalties, even when the player fails to hit a mole in the allocated time: no points are lost, the user still has opportunities to repeat and succeed. In this vein, we decided to take advantage of recent wireless HMD technologies and hand tracking capabilities, thus providing an easier to deploy and use setup. We selected the Oculus Quest 2, since it offers inside-out markerless tracking without additional hardware, requires little environment setup and doesn't need a computer to run. This freedom makes the device ideal and easy to setup for non-technical users. Plus, it is cheaper when compared to existing HMDs. Development occurred using Unity and Oculus SDK, based on C# scripts.

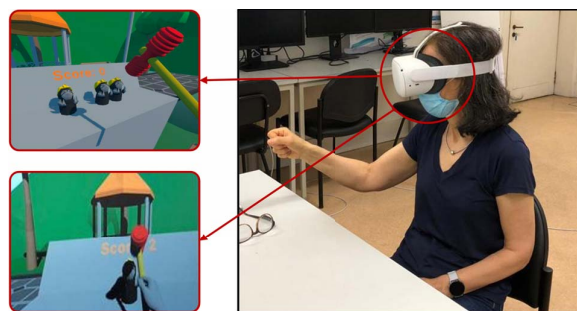


Figure 2: Example of a participant using a HMD to experience the whac-a-mole serious game, in particular, using her right arm to grab a virtual hammer and hit predefined targets. A movement predetermined as very useful for upper-limb stroke survivors physical rehabilitation.

4 USER STUDY AND RESULTS

A user study was conducted to evaluate the usability of the prototype. We focused on two experimental conditions: C1 - normal setting and C2 - mirror setting. The goal was to use a virtual toy hammer to hit 15 targeted objects appearing in random positions. Participants' opinion was gathered through a post-task questionnaire and interview including: demographic information, the System Usability Scale questionnaire (SUS), and questions concerning user preferences. Participants were instructed on the experimental setup, the task, and gave their informed consent. At the end, participants answered the post-task questionnaire and a small interview was conducted to understand their opinion towards the proposed prototype. This study has obtained ethics approval.

We recruited 10 participants (3 female) including Researchers, Faculty Members, MSc and PhD students having previous experience evaluating VR-based applications. Using condition C1, participants were quicker to grab the virtual hammer and hit the targets. They described the game as fun and active. Movements were easy to understand and to perform. This may be caused by the technology

used, which allowed tracking the hands without additional sensors or commands. Participants were confident other individuals would find it easy to use without further explanation from technical persons, which is illustrated by the SUS result of 87.81. As for condition C2, participants emphasize the adaptation period to the mirror mode was higher, as illustrated by the SUS score of 65.94. Although they did not find this mode unnecessarily complex or inconsistent, they recognize a greater time period and mental effort was required. Yet, they improved after initial disorientation with the symmetrical movement, which ponders the question if survivors already familiar with mirror therapy would have a better performance. Some participants reported the integration of customization features, namely time options limiting the game to 2, 5 or 10 minutes. They also recommend using more rows of moles, better mimicking the original game.

5 CONCLUDING REMARKS AND FUTURE WORK

We proposed an immersive VR prototype for upper-limb post-stroke physical rehabilitation. As for the user study, condition C1 rated higher in terms of learnability and ease of use. Yet, no conclusions about their efficacy in rehabilitation can be made at this time. Further testing with the target audience was not possible due to the effects of the global pandemic. This showed the need of VR rehabilitation solutions at real-life settings, in order to avoid the absence of these activities, which are essential to provide proper care to the survivors.

This study is being expanded by conducting a longitudinal study (3 months) with stroke survivors. This effort may help inform the next steps of our research, in particular how to improve our prototype and advance innovation in stroke rehabilitation for home settings. We will also move into the integration of customization features, allowing the use of various serious games and different levels of difficulty and rewards, aiming to elicit greater interest by the stroke survivors over time, when considered to existing alternatives.

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